INDIAN STATISTICAL INSTITUTE, KOLKATA | COURSE PROJECT: REGRESSION TECHNIQUES

Flexibility vs. Interpretability in High-Dimensional Cancer Classification

Regularized Regression, Tree Ensembles, and Neural Networks on RNA-Seq Data ($p \gg n$)

Rajveer Vora | **Amartya Amritanshu** | **P. Sanjith Reddy**

November 2026

The High-Dimensional Setting: $p \gg n$ Transcriptomics

Theoretical breakdowns and biological collinearity in bulk RNA-Seq

Extreme Dimensionality

- $n = 801$ patient samples across $K = 5$ cancer types.
- $p = 20,531$ continuous gene expression features.
- Underdetermined system: features outnumber observations $25:1$.

Singular Covariance

- Covariance $\mathbf{S} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X} \in \mathbb{R}^{p \times p}$.
- $\text{rank}(\mathbf{S}) \leq n - 1 = 800 \ll 20,531$.
- $\geq 19,730$ zero eigenvalues.
- **OLS and Fisher LDA fail** due to singular non-invertibility.

Biological Collinearity

- Genes act in co-regulated pathways and transcriptional modules.
- Thousands of gene pairs have Pearson $|r| > 0.85$.
- Infinite hyperplanes achieve perfect linear training separation.

Takeaway: Classical unregularized estimators fail mathematically; the challenge is identifying stable, interpretable biomarkers without overfitting.

SECTION 01

Dataset and Exploratory Analysis

RNA-Seq distributions, variance screening, and inherent linear separability

Preprocessing and Unsupervised Variance Screening

Transforming raw read counts and filtering non-informative variation

Variance Stabilization

Distributional Skewness:

- Raw counts span 0 to 10^6.
- Log-transformation:

$$z_{ij} = \log_2(x_{ij} + 1)$$

- Symmetrizes distributions and moderates outlier spikes.

Top 1,000 Variance Rationale

Three Essential Factors:

- **Noise removal**: Housekeeping genes.
- **Feasibility**: Fast iterative solving.
- **Empirical saturation**: $p \in [500, 1000]$ yields > 0.99 Macro-F1.

Robust Screening Metric

Median Absolute Deviation (MAD):

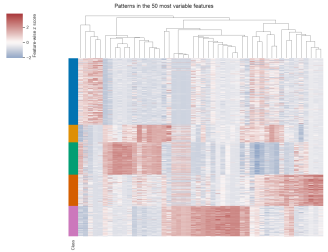
$$\text{MAD}_j = \text{med}_i |z_{ij} - \text{med}_{i'}(z_{i'j})|$$

- Robust against count spikes.
- Computed solely on training folds.

Takeaway: Unsupervised MAD screening removes static features without viewing labels, preserving strict test set independence.

Clustered Heatmap: High-Variance Transcriptional Profiles

Unsupervised clustering reveals strong class-specific expression signatures



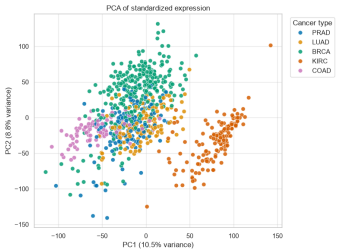
Pattern Observations

- **Top 50 Variable Genes**: Standardized relative expression across all 801 patients.
- **Block-Diagonal Structure**: Distinct expression signatures align clearly with cancer cohorts (BRCA, KIRC, COAD, LUAD, PRAD).
- **Hierarchical Dendrogram**: Patient samples cluster predominantly by tissue origin.
- **Collinear Blocks**: Correlated gene blocks reflect co-regulated biological pathways.

Takeaway: Natural clustering validates that the dataset contains strong class-specific signals, motivating linear classification.

Principal Component Analysis: 2D Unsupervised Projection

Spectral decomposition confirms natural separability in orthogonal space



Variance Decomposition

- **PC1 (18.6% variance)**: Separates breast carcinoma (BRCA) cleanly from kidney renal clear cell (KIRC).
- **PC2 (12.3% variance)**: Distinguishes colon adenocarcinoma (COAD) and lung (LUAD).
- **Cumulative Variance**: **30.9%** of total variation captured in just two linear dimensions.
- **Cluster Separation**: Cohorts form distinct clusters without non-linear transformations.

Takeaway: The 5 tumor types are largely linearly separable in an unsupervised 2D projection, bounding the need for complex classifiers.

SECTION 02

Statistical Methodology

Regularized multinomial regression, tree ensembles, and neural baselines

Regularized Linear Models: Ridge, Lasso, and Elastic Net

Controlling ill-conditioned multinomial cross-entropy in $p \gg n$

Ridge (L_2 Regularization)

$$\min_{\mathbf{W}} \mathcal{L}_{\text{CE}} + \frac{\lambda}{2} \sum_k \|\mathbf{w}_k\|_2^2$$

- Strictly convex Hessian ($\lambda \mathbf{I}$).
- Smooth weight shrinkage.
- Retains all **1,000 features**.
- Optimal probability calibration.

Lasso (L_1 Regularization)

$$\min_{\mathbf{W}} \mathcal{L}_{\text{CE}} + \lambda \sum_k \|\mathbf{w}_k\|_1$$

- Polyhedral L_1 geometry.
- Highly sparse: **89 active genes**.
- **Limitation:** Arbitrarily selects 1 gene from collinear clusters.

Elastic Net ($L_1 + L_2$)

$$\min_{\mathbf{W}} \mathcal{L}_{\text{CE}} + \lambda \sum_k \left[\alpha \|\mathbf{w}_k\|_1 + \frac{1-\alpha}{2} \|\mathbf{w}_k\|_2^2 \right]$$

- Sparsity + stability ($\alpha = 0.5$).
- Selects **218 active genes**.
- **Grouping property:** Retains co-regulated pathway genes.

Takeaway: Regularization prevents weight divergence on separable data; Elastic Net addresses collinear feature selection directly.

The Grouping Effect and Non-Linear Baselines

Theoretical bound under collinearity and comparison against tree / neural models

Elastic Net Grouping Property Theorem

For two standardized features $\mathbf{x}_i, \mathbf{x}_j$ with sample correlation r:

$$|\hat{w}_i - \hat{w}_j| \leq \frac{\sqrt{2(1-r)}}{\lambda(1-\alpha)}\|\mathbf{y}\|_2$$

- As correlation $r \to 1$, coefficient difference bound $|\hat{w}_i - \hat{w}_j| \to 0$.
- Strongly correlated genes enter or leave the active set together.
- Resolves Lasso's arbitrary feature dropping across biological pathways.

Non-Linear Baseline Architectures

Tree Ensembles:

- **Random Forest**: 200 bagged trees, feature subsampling $m = \lfloor \sqrt{p} \rfloor \approx 31$.
- **XGBoost**: Gradient boosting ($\eta = 0.10$, max depth 4, 100 rounds).

Neural Networks:

- **PCA-MLP**: 434 PCs (95% variance) $\to 128 \to 64 \to 5$.
- **Sparse NN**: Input group-lasso penalty $\lambda \sum \|\mathbf{W}^{(1)}_{:j}\|_2$.

Takeaway: Elastic Net guarantees coefficient grouping for correlated genes, whereas greedy tree splits choose arbitrarily.

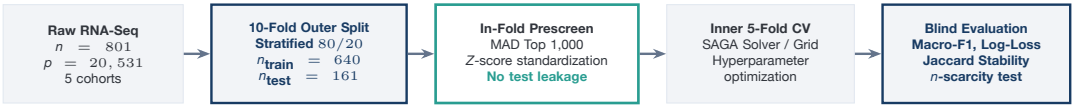
SECTION 03

Experimental Design

Strict validation protocol, stability metrics, and stress tests

Experimental Pipeline: Zero-Leakage Validation

Strict nested evaluation protocol across 10 independent stratified splits



Strict Test Set Isolation

Feature screening and standardization parameters (μ, σ) are computed solely on training folds; test folds remain completely blind until final evaluation.

Repeated Outer Splits

10 independent random resamplings quantify performance variance, feature selection stability, and model robustness under varied sample draws.

Takeaway: Nested protocol ensures that neither feature screening nor parameter scaling leaks test set information.

Multi-Criteria Evaluation: Beyond Classification Accuracy

Evaluating discrimination, probabilistic calibration, and biomarker stability

Macro-F1 Score

$$\text{Macro-F1} = \frac{1}{K}\sum_{k=1}^{5}\frac{2P_kR_k}{P_k + R_k}$$

- Unweighted class mean.
- Prevents BRCA ($n = 300$) from masking COAD ($n = 78$).

Multi-Class Log-Loss

$$\mathcal{L}_{\log} = -\frac{1}{N}\sum_{i,k} y_{ik}\log(\hat{p}_{ik})$$

- Tests probability calibration.
- Penalizes overconfident wrong assignments.

Selection Stability

$$D = \frac{2|S_a \cap S_b|}{|S_a| + |S_b|},\ J = \frac{|S_a \cap S_b|}{|S_a \cup S_b|}$$

- Pairwise split overlap.
- Quantifies biomarker reproducibility.

Takeaway: In linearly separable settings, accuracy saturates; log-loss and stability expose genuine model quality.

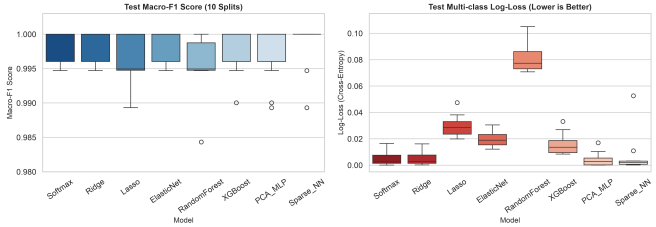
SECTION 04

Results and Comparison

Macro-F1 saturation, probability calibration, and the Pareto frontier

Benchmark Performance Across 10 Repeated Splits

Macro-F1 saturation vs. prominent divergence in probabilistic log-loss



Quantitative Breakdown

- **Macro-F1 Saturation**: All 8 models achieve mean Macro-F1 ≥ 0.993. Accuracy alone cannot rank these architectures.
- **Log-Loss Divergence**:
  - Linear models and PCA-MLP achieve < 0.020 log-loss (sharp calibration).
  - Random Forest exhibits 0.081 **log-loss** due to probability softening from vote averaging.
  - Sparse NN yields 0.063 **log-loss**.

Takeaway: Multi-class log-loss reveals calibration deficits in tree ensembles that are invisible in accuracy metrics.

Comprehensive Model Comparison Table

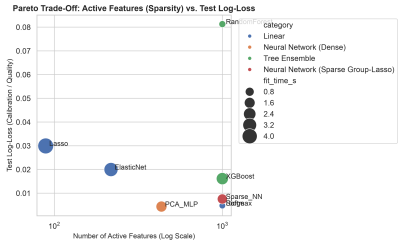
Evaluation across discrimination, calibration, active sparsity, and runtime

Model	Macro-F1	Log-Loss	Active Features	Fit Time (s)	Stability (D / J)
Softmax	0.998 ± 0.003	0.005 ± 0.002	$1,000$	0.004	1.000 / 1.000
Ridge (L_2)	0.998 ± 0.003	0.005 ± 0.002	$1,000$	0.004	1.000 / 1.000
Lasso (L_1)	0.995 ± 0.006	0.030 ± 0.015	**89**	4.44	0.609 / 0.443
Elastic Net	**0.998** ± 0.003	0.020 ± 0.007	**218**	3.65	**0.725** / **0.573**
Random Forest	0.994 ± 0.006	0.081 ± 0.010	$1,000$	0.13	0.276 / 0.160
XGBoost	0.995 ± 0.006	0.028 ± 0.009	$1,000$	2.18	0.443 / 0.285
PCA-MLP	0.998 ± 0.003	**0.004** ± 0.001	434	1.68	N/A
Sparse NN	0.997 ± 0.004	0.063 ± 0.015	$1,000$	1.25	N/A

Takeaway: Elastic Net matches the top F1 score (0.998) while eliminating 78% of candidate genes, outperforming trees in stability and calibration.

Active Feature Sparsity vs. Log-Loss: The Pareto Frontier

Evaluating the direct trade-off between model parsimony and probability calibration



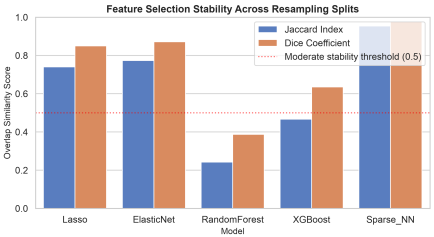
Trade-off Spectrum

- **Dense Boundary**: Ridge and PCA-MLP achieve minimal log-loss (0.004–0.005) by retaining all features or 434 principal components.
- **High Compression**: Lasso compresses to **89 active genes** with a modest log-loss trade-off (0.030).
- **The Optimal Compromise**: **Elastic Net** achieves 0.020 log-loss with **218 genes**, balancing parsimony and calibration.
- **Tree Ensembles**: Fall far from the frontier due to softened probabilities.

Takeaway: Elastic Net establishes the best compromise between biological interpretability and sharp test calibration.

Feature Selection Stability across Resampling Splits

Evaluating biomarker reproducibility under high pairwise collinearity



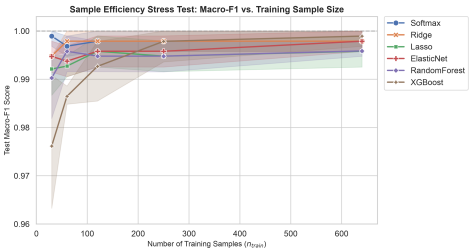
Stability Findings

- **Elastic Net Dominance**: Achieves top stability among sparse models (**Dice 0.725**, **Jaccard 0.573**) via L_2 grouping.
- **Lasso Degradation**: Moderate stability ($D = 0.609$, $J = 0.443$); randomly selects 1 feature from collinear groups.
- **Tree Split Collapse**: Random Forest ($J = 0.160$) and XGBoost ($J = 0.285$) show severe selection instability.
- **Mechanism**: Collinear features offer nearly identical impurity drops, causing tree splits to fluctuate randomly.

Takeaway: Biomarker discovery requires L_2 regularization; greedy tree splitters cannot stably select correlated genes.

Sample Efficiency Stress Test: $n_{\text{train}} = 30 \rightarrow 640$

Simulating extreme clinical sample scarcity down to 6 patients per cancer cohort



Scarcity Analysis

- **Linear Resilience**: Regularized linear models (Ridge, Softmax, Elastic Net) remain remarkably robust ($\geq \mathbf{0.995}$ Macro-F1) even at $n_{\text{train}} = 30$.
- **Tree Ensemble Drop**: XGBoost degrades to 0.976 ± 0.018 (minimum fold score 0.954). Random Forest drops to 0.985.
- **Inductive Bias**: Strong linear assumptions protect against small-sample overfitting; non-linear step boundaries overfit when sample density is low.

Takeaway: Regularized linear models exhibit superior sample efficiency, retaining high performance even under extreme sample scarcity.

SECTION 05

Interpretation and Conclusion

Answers to research questions, limitations, and practical recommendations

Answers to Core Research Questions

Synthesized conclusions from benchmark and stress-test experiments

Q1. Flexibility in $p \gg n$?

No Advantage.

- Linear separability means deep nets and boosting offer no accuracy gain over linear models.
- Non-linear models inflate calibration error.

Q2. Collinearity Impact?

Destabilizes Trees & Lasso.

- Lasso arbitrarily drops pathway co-regulators.
- Trees suffer split instability ($J = 0.160$).
- Elastic Net L_2 grouping stabilizes selection ($J = 0.573$).

Q3. Optimal Architecture?

Elastic Net.

- Matches 0.998 F1 with 78% fewer genes (218 active).
- Balances calibration (0.020) and biomarker stability.
- Robust under sample scarcity ($n = 30$).

Takeaway: In high-dimensional transcriptomics, inductive bias and regularization dominate architectural complexity.

Honest Limitations and Methodological Scope

Distinguishing benchmark findings from clinical translation requirements

Single-Source Cohort

- Single TCGA batch ($n = 801$).
- Multi-center batch variations not assessed.
- Requires external prospective cohort validation.

Primary vs. Fine Subtypes

- Primary tissue sites have large expression divergence.
- Intra-organ subtypes (e.g. breast luminal A vs. B) present subtle non-linear boundaries.

Clinical Translation

- 218 genes form a computational screen for diagnostic assays.
- Candidates require biological assay validation before claiming causal biomarker status.

Takeaway: Findings demonstrate strong linear separability for primary cancer types, while subtle subtyping remains an open question.

Summary and Practical Recommendations

Key guidelines for practitioners analyzing high-dimensional RNA-Seq data

Core Recommendations for Practitioners

- **Default to Elastic Net**: Use Elastic Net ($\alpha \approx 0.5$) as primary baseline; it combines high accuracy, 78% sparsity, and grouping stability.
- **Avoid Unregularized Models**: Never fit standard OLS or LDA on $p \gg n$ data due to singular covariance.
- **Evaluate Beyond Accuracy**: Rely on multi-class log-loss and stability metrics to detect calibration loss and biomarker instability.
- **Exercise Caution with Trees**: Do not rely on tree split importance under high gene collinearity.

Takeaway: Regularized linear models provide the most reliable, calibrated, and reproducible foundation for high-dimensional cancer classification.

Project Artifacts & Code

- **Dataset**: 801 TCGA RNA-Seq samples ($K = 5$).
- **Pipeline**: 10-fold nested CV with zero test leakage.
- **Report**: 10-page academic project paper with full derivations.

Thank You | Questions & Discussion